

IS6400 Business Data Analytics

Dr. LIU Junming

City University of Hong Kong

Lecture 7

Reading Material:

- ☐ Introduction to time series

- ☐ Quantitative Forecasting

 - ☐ Statistical Methods

 - ☐ Regression Analysis

Introduction

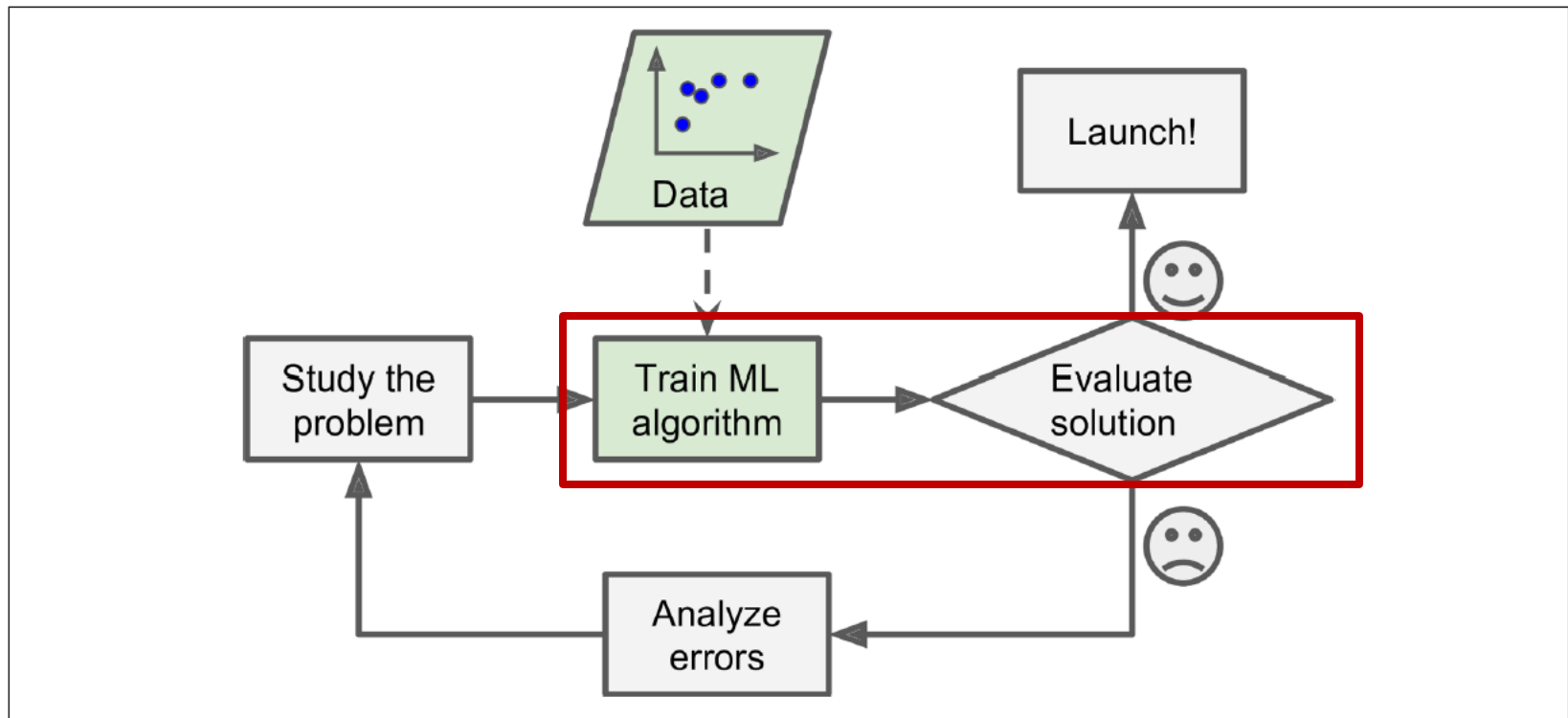


Figure 1-2. Machine Learning approach

Review – Linear Regression

- ❖ Identify what to be predicted and potential predictors

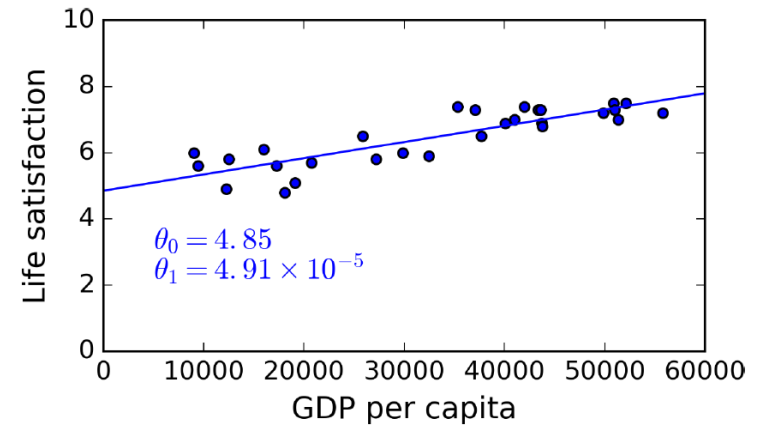
Statisticians say	Data Miners say	Social Scientists say	They all Mean
Independent Variables	Predictor Variables	Explanatory Variables	Stuff you can use to predict the target
Dependent Variable	Target Variable	Response Variables	The thing you want to predict

Review – Linear Regression

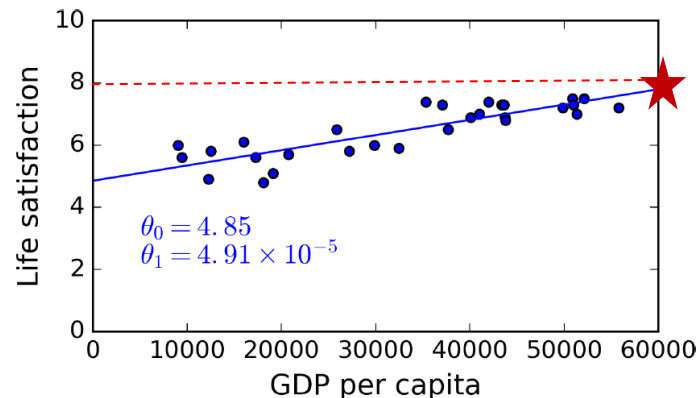
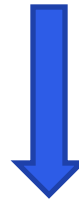
Table 1-1. Does money make people happier?

Country	GDP per capita (USD)	Life satisfaction
Hungary	12,240	4.9
Korea	27,195	5.8
France	37,675	6.5
Australia	50,962	7.3
United States	55,805	7.2

Train

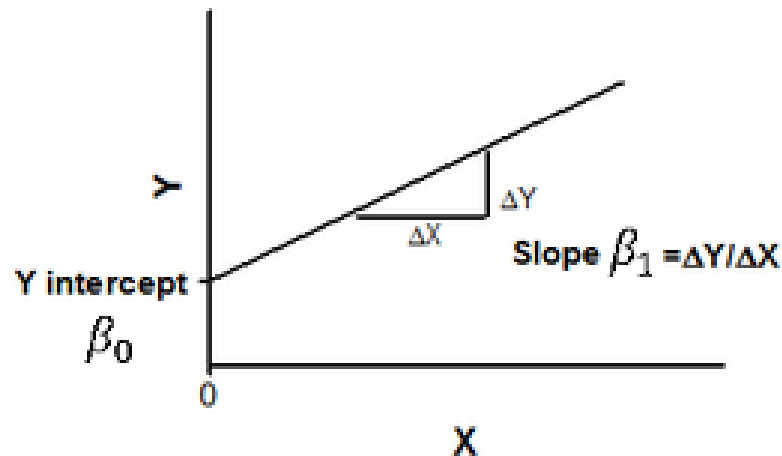


Implement



Review – Linear Regression

- ❖ To predict the value of one dependent variable on the basis of another independent variable
 - Dependent Variable: **Y**
 - Independent Variable: **X**



Review – Regression Tree

Machine Learning Model: Decision Tree Regression

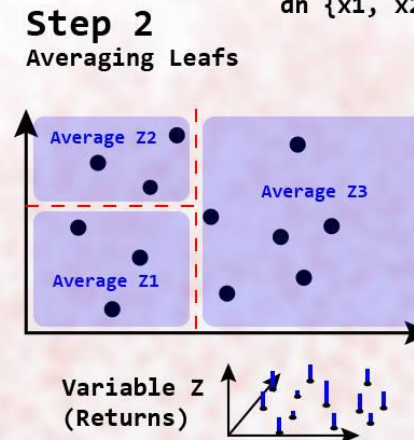
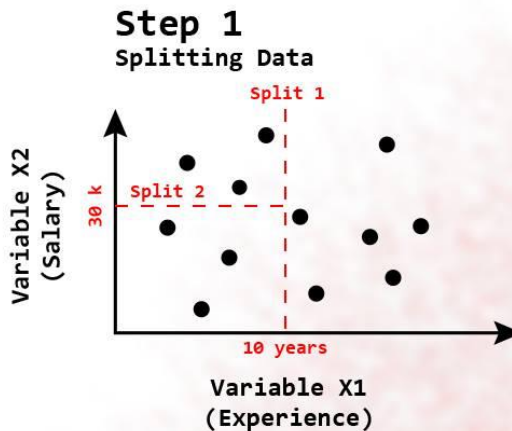
Training

Data:

$d1 \{x1, x2, z\}$

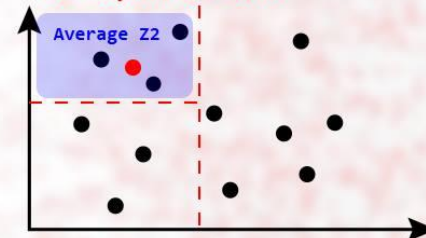
...

$dn \{x1, x2, z\}$

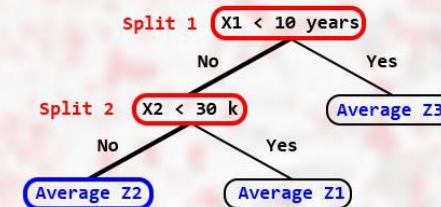


Step 3
Introducing New Data

Introducing New Data Point
 $X1 = 6 \text{ years}$ $X2 = 40 \text{ k}$



Step 4
Finding Z using Decision Tree

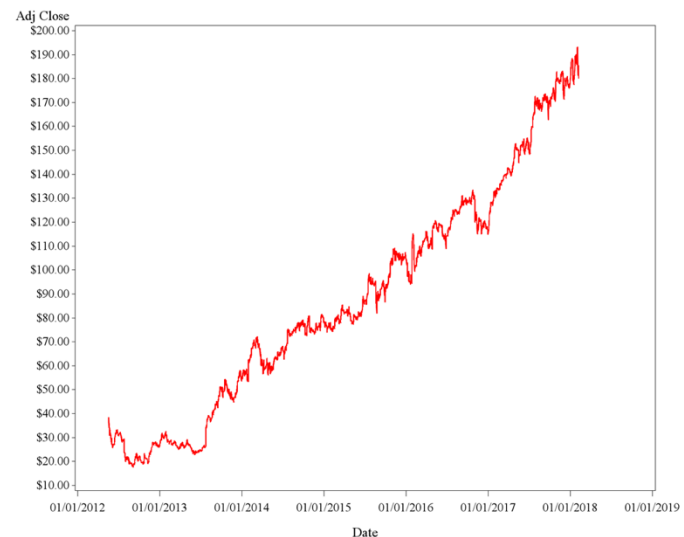
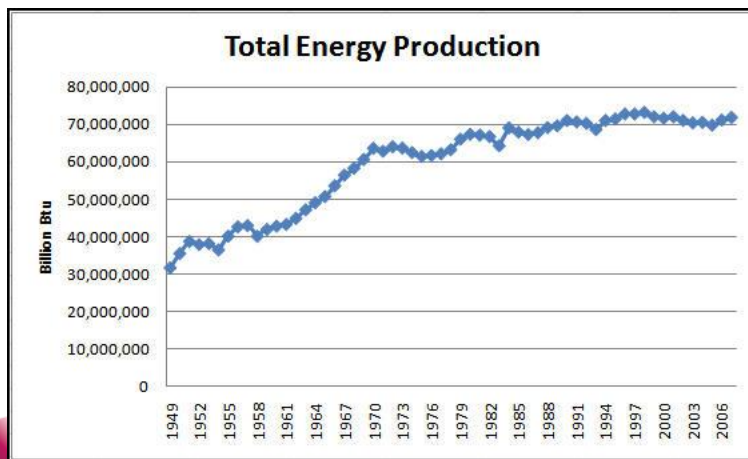


Result: The the Z value of the new data point ● is Z2!

Introduction – Time Series

A **Time Series** is a set of observations of a variable at regular time intervals, such as yearly, monthly, weekly, daily, 15 minutes interval, etc.

- ❖ To predict the value of one dependent variable on the basis of **time information**
 - Time indexed Dependent Variable: **Y_t**
 - Time indexed Independent Variable: **X_t**



Stock price of Facebook

Introduction – Time Series

❖ What will happen NEXT?



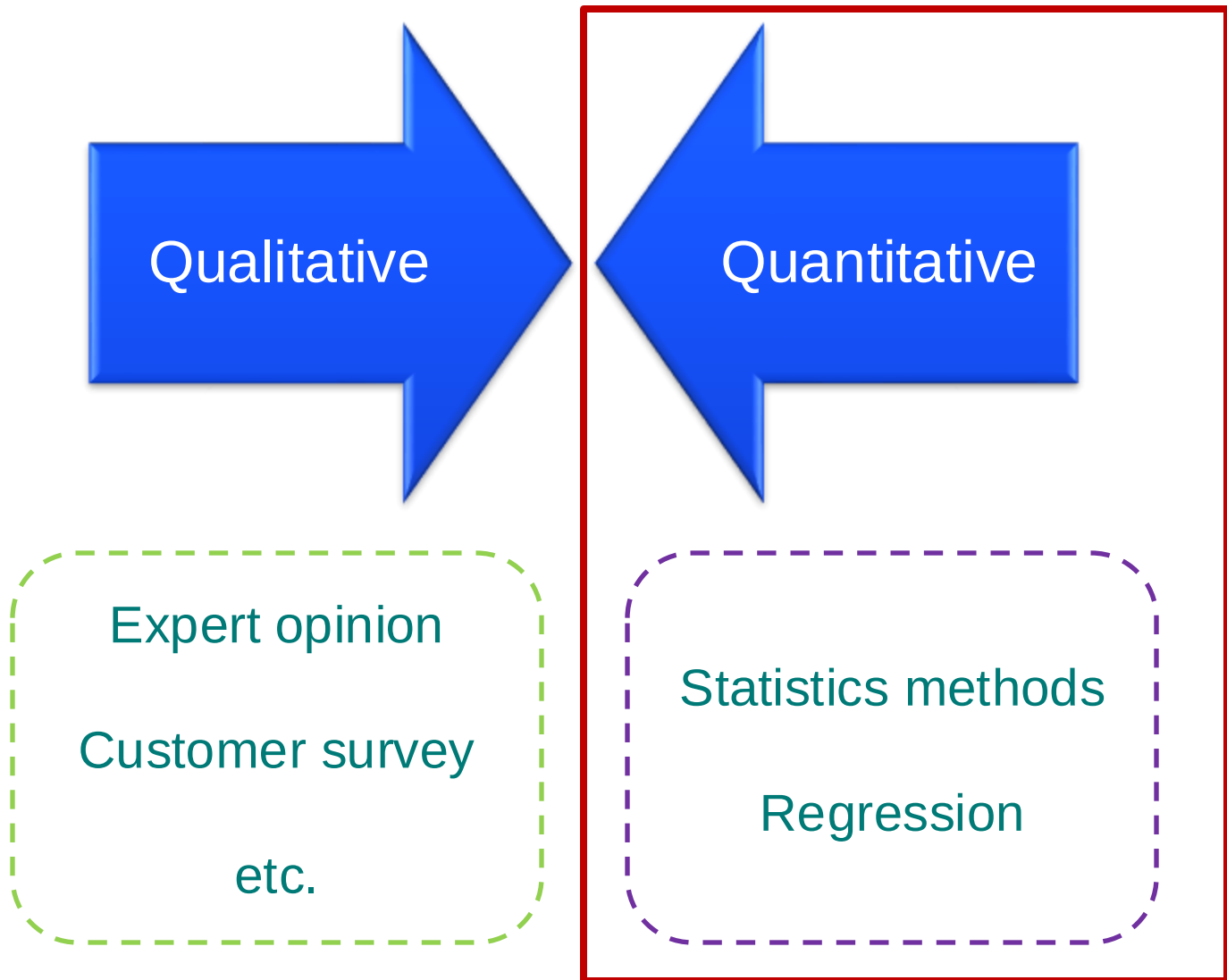
Time series data



Time series prediction



Time series prediction



Forecasting Techniques

❖ Qualitative and judgmental

- ❧ Historical analogy – comparative analysis with a previous situation
- ❧ Delphi Method – response to a sequence of questionnaires by a panel of experts
- ❧ Market Research

❖ Quantitative

- ❧ Extrapolation (or time series) methods
- ❧ Econometric (or causal) methods

Qualitative and Judgmental Forecasting

- ❖ Qualitative techniques rely on experience and intuition.
- ❖ They are necessary when historical data is not available or when predictions are needed far into the future.
- ❖ The historical analogy approach obtains forecasts through comparative analysis with prior situations.

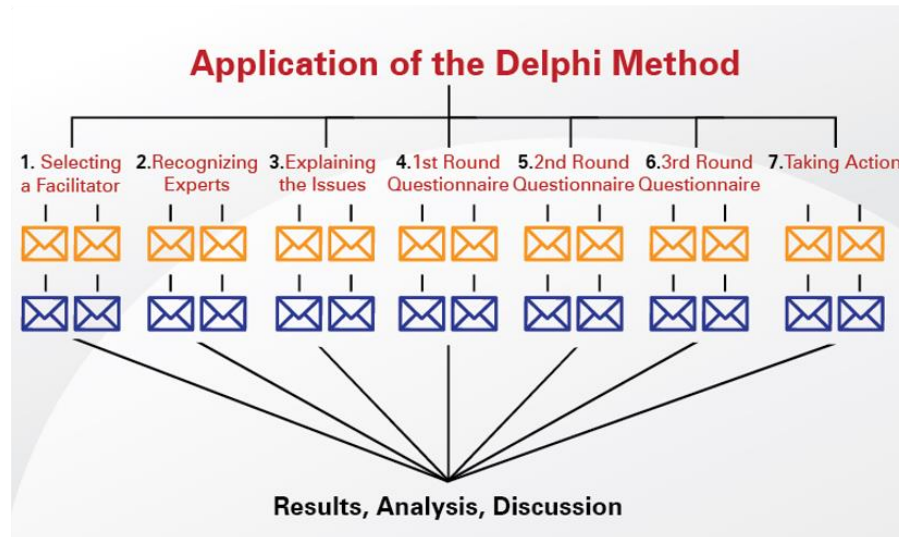


Qualitative and Judgmental Forecasting

❖ Qualitative and Judgmental Forecasting

- **The Delphi method**

Use a panel of experts to reach a convergence of opinion on the forecasted variable.



Quantitative Forecasting

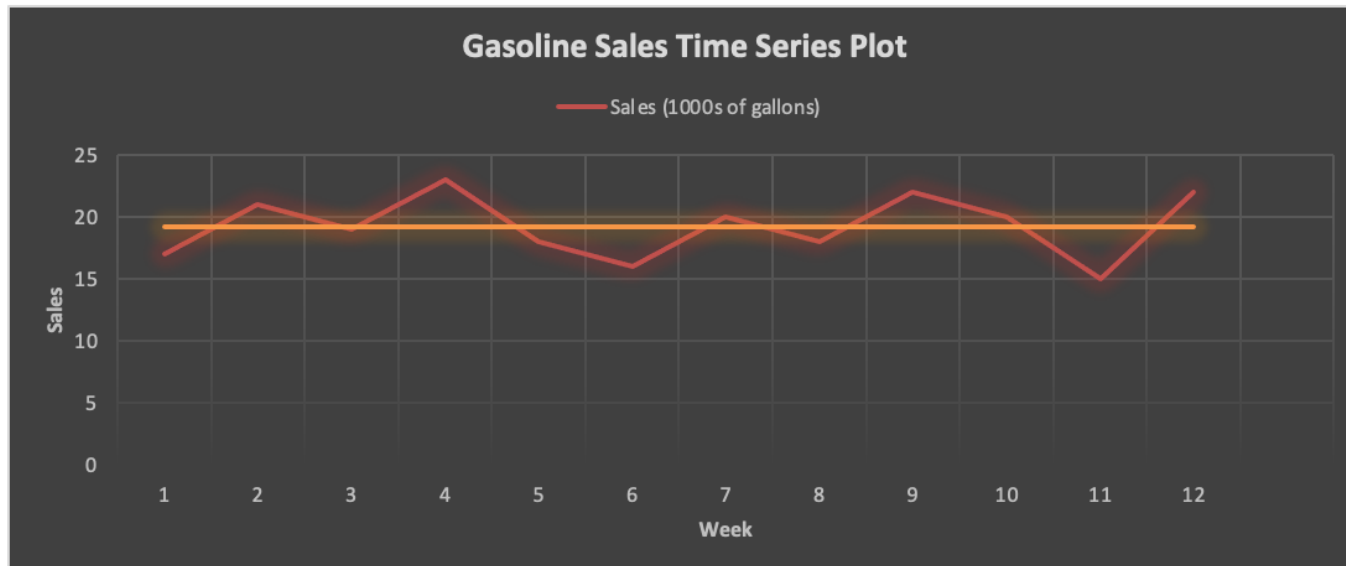
❖ Quantitative methods

- ↻ Search for patterns or relationships in historical data and then make forecasts. **Using past to predict future.**
- It is not always easy to uncover historical patterns or relationships.
 - It is often difficult to separate the noise, or random behavior, from the underlying patterns.
 - Some forecasts may attribute importance to patterns that are in fact random variations and are unlikely to repeat themselves.
- There are no guarantees that past patterns will continue in the future.
 - History does not always repeat itself. Paradigm shifting
 - A good forecasting model is the one that fits the historical data well. There is no guarantee that it will predict the future accurately.

Time Series Patterns

❖ Horizontal Pattern

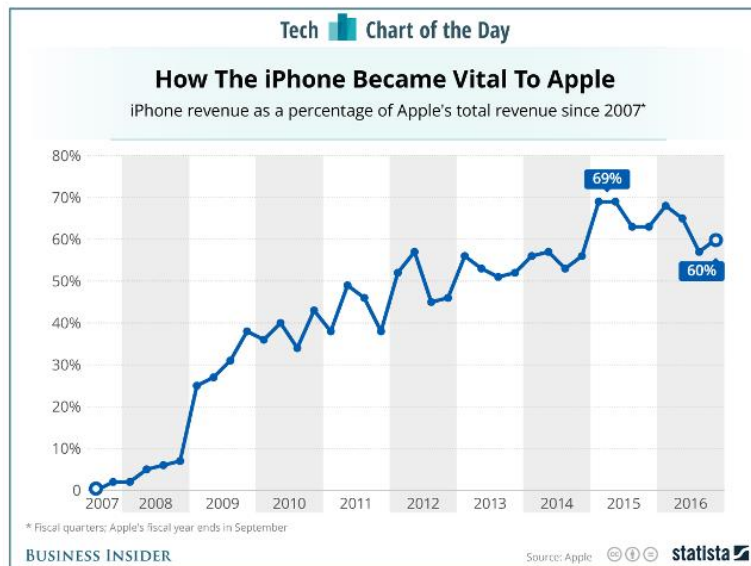
- A **horizontal pattern** exists when the data fluctuate around a constant mean.
- Generally, regression and convergence



Time Series Patterns

❖ Trend Pattern

- **Trend pattern** exists when the time series data shifts to relatively higher or lower values over a longer period of time.



Time Series Patterns

❖ Seasonal Pattern

- **Seasonal pattern** exists when time series data exhibits a repeating pattern over a (fixed) period (e.g., yearly) due to seasonal influences.



Time Series Patterns

❖ Seasonal + Trend

- A combination of seasonal Pattern and Trend.

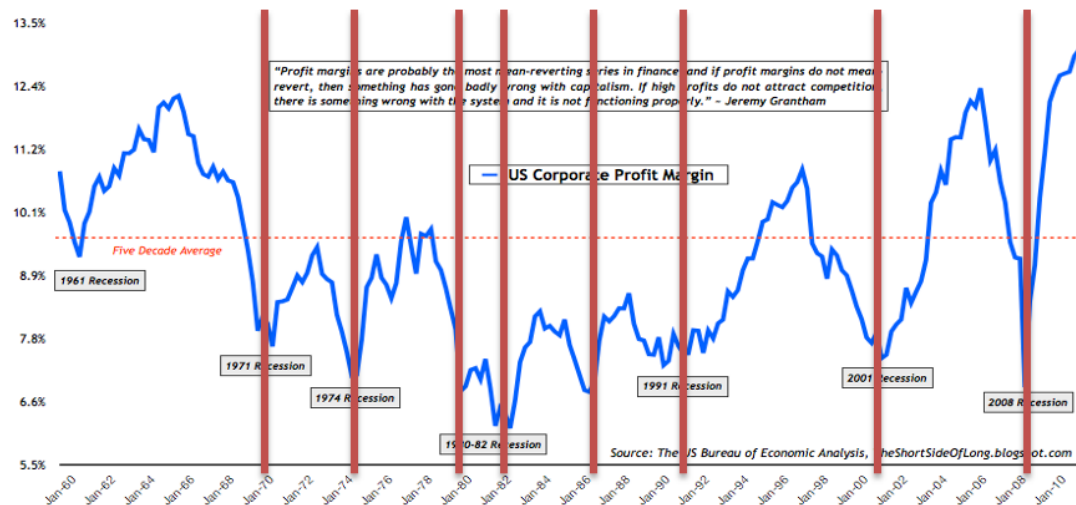


<http://calculatedrisk.blogspot.com/>

Time Series Patterns

❖ Cyclical pattern

- Alternative sequence of points below and above the trend line. Cyclical pattern is hard to identify empirically.
- Change a word, in a cyclical pattern, you can find repeated pattern, but the time period varies. Seasonal pattern is a special case.



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- ☒ **Quantitative Forecasting**
 - ☒ **Statistical Methods**
 - ☐ Regression Analysis

Time Series Component

- ❖ The different components reflect the impact of different factors. The overall time series is the combination of different components.
- ❖ Additive Model

$$Y_t = T_t + S_t + C_t + I_t$$

- ❖ Multiplicative model (linear in log form)

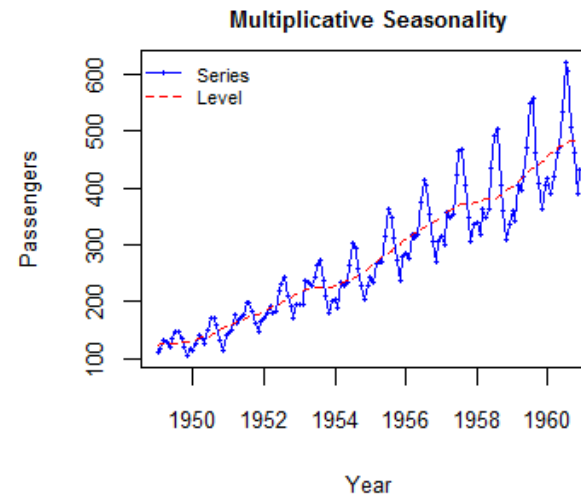
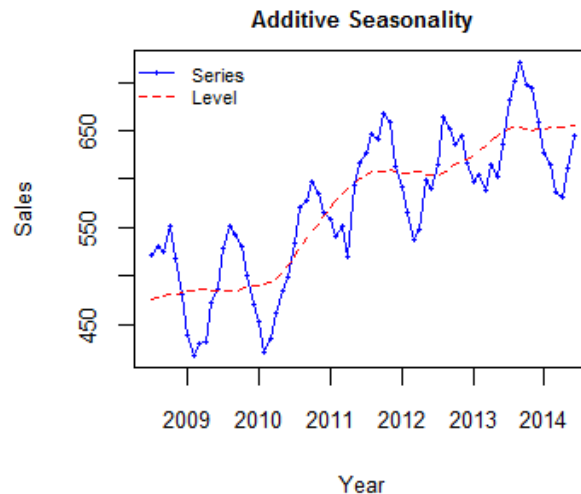
$$Y_t = T_t S_t C_t I_t$$

where

- T_t = Trend value at period t
- S_t = Seasonality value for period t
- C_t = Cyclical value at time t
- I_t = Irregular (random) value for period t

Additive vs. Multiplicative Seasonal Model

- ❖ In an **additive seasonal model**, an appropriate seasonal index is added to a base forecast.
- ❖ In a **multiplicative seasonal model**, a base forecast is multiplied by an appropriate seasonal index.
- ❖ We focus on multiplicative seasonal model.

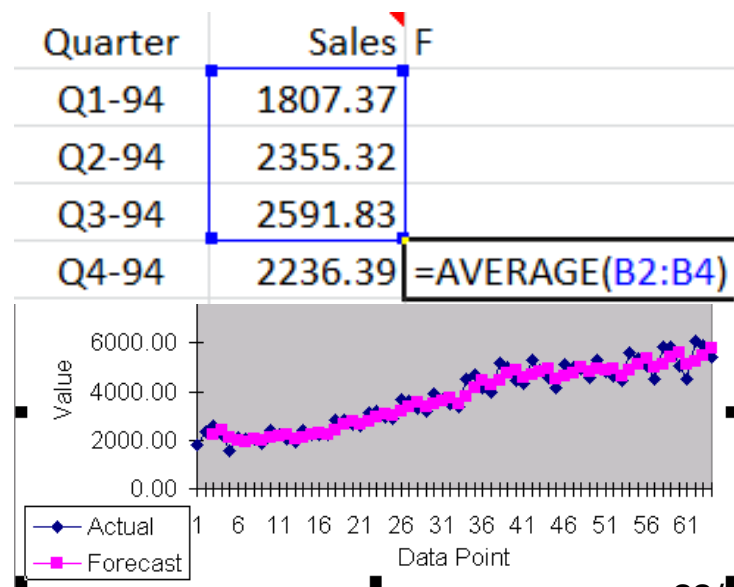


Simple Moving Average

- ❖ *Average random fluctuations in a time series* to infer short-term changes in direction
- ❖ **Assumption**: future observations will be similar to recent past
- ❖ Forecast = average of most recent k observations

$$F_{t+1} = \frac{Y_t + \dots + Y_{t-m+1}}{m}$$

- ❖ So, how do you decide m?
What if m=1 or 2?



Weighted Moving Average

- ❖ A natural extension to moving average.
 - ⌘ Weight the most recent k observations, with weights that add to 1.0
 - ⌘ *Higher weights on more recent observations generally provide more responsive forecasts to rapidly changing time series*
 - ⌘ Rational: recent observations are more similar to what will happen next

$$F_{t+1} = \alpha_1 Y_t + \dots + \alpha_k Y_{t-m+1}$$

$$\alpha_1 + \dots + \alpha_m = 1$$

Exponential Smoothing

- ❖ So: an easier approach (with one parameter)

- ∞ Weights decay exponentially

- ∞ Sum of weights approach 1

$$F_{t+1} = \alpha Y_t + \alpha(1 - \alpha)Y_{t-1} + \alpha(1 - \alpha)^2 Y_{t-2} + \alpha(1 - \alpha)^3 Y_{t-3} + \dots$$

- ❖ Exponential smoothing model:

$$\begin{aligned} F_{t+1} &= L_t = (1 - \alpha)L_{t-1} + \alpha Y_t \\ &= L_{t-1} + \alpha(Y_t - L_{t-1}) \end{aligned}$$

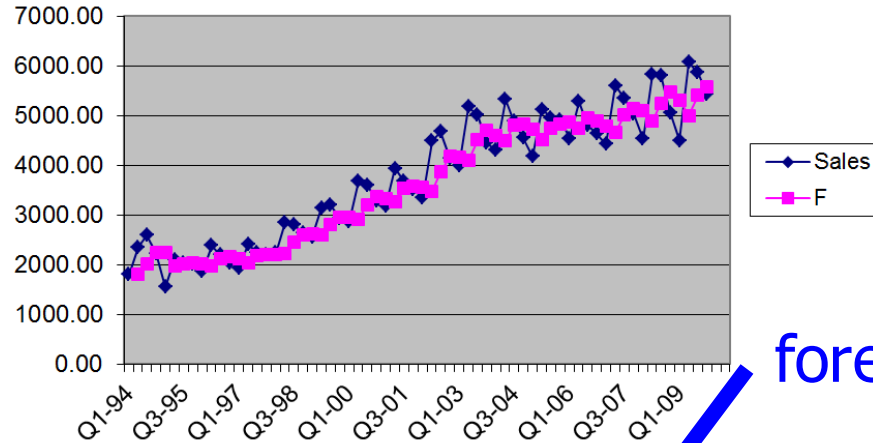
last prediction + adjustment on error

- ❖ F_{t+1} is the forecast for time period $t+1$,
- ❖ L_t is the level for period t , ($L_1 = Y_1$)
- ❖ Y_t is the observed value in period t , and
 α is a constant between 0 and 1, called the smoothing constant.

Higher smoothing constant (higher weights on more recent observations) provide forecasts that respond more rapidly to changes in series.

Soft Drink Sales

Quarter	Sales	L	F	e	Alpha	
Q1-94	1807.37		1807.37			0.38286
Q2-94	2355.32		2017.16	1807.37	547.95	RMSE 470.5061
Q3-94	2591.83	$=\$G\$1*\$B4+(1-\$G\$1)*\$C3$	2017.16	574.67	Avg_E	154.26
Q4-94	2236.39		2236.88	2237.18	-0.79	
Q1-95	1549.14		1973.57	2236.88	-687.74	
Q2-95	2105.79		2024.19	1973.57	132.22	



Q3-09	5868.67		5587.56	5413.16	455.51	
Q4-09	5432.24		5528.09	5587.56	-155.32	
Q1-10				$=C65$		

Quiz

	Date	Sales
0	1/1/2020	266.0
1	2/1/2020	145.9
2	3/1/2020	183.1
3	4/1/2020	119.3
4	5/1/2020	180.3
5	6/1/2020	168.5
6	7/1/2020	231.8
7	8/1/2020	224.5
8	9/1/2020	192.8
9	10/1/2020	122.9
10	11/1/2020	336.5

Q1: if we use a moving average with a window size of 5, what is the predicted value for the demand on 11/1/2020?

Q2: if we use exponential smoothing with $\alpha=0.2$, what is the predicted value for the demand on 3/1/2020?

Double Moving Average

- ❖ DMV: This method is best for historical data with a trend but no seasonality.

$$M_t = [Y_{t-n+1} + Y_{t-n+2} + \dots Y_{t+1}]/n$$

$$D_t = [M_{t-n+1} + M_{t-n+2} + \dots M_{t+1}]/n$$

$$L_t = M_t + (M_t - D_t)$$

$$T_t = 2*(M_t - D_t)/(n-1)$$

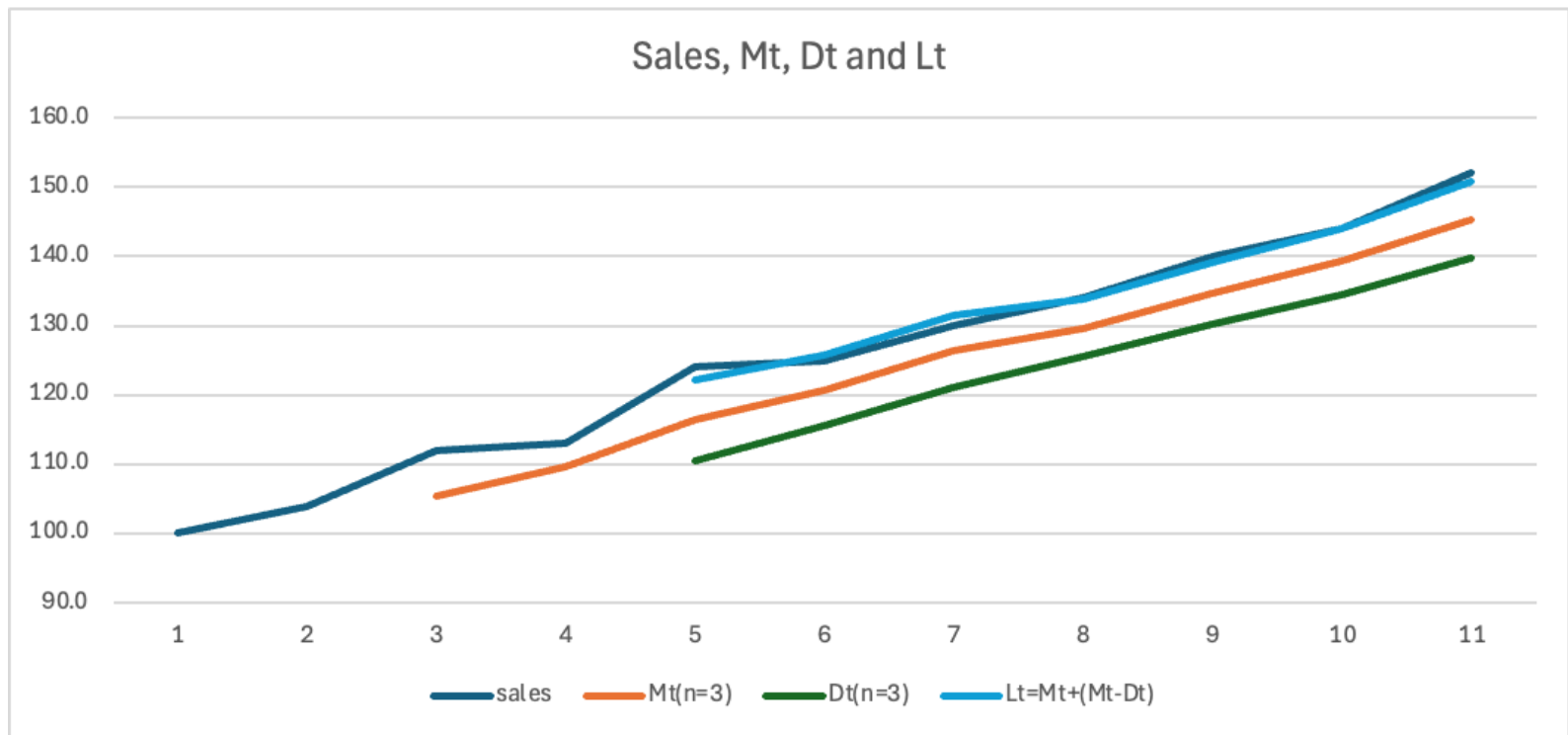
Double moving average to smooth level and trend

Use L_T and T_T in the linear trend equation to forecast k periods beyond period T :

$$F_{t+k} = L_t + T_t k$$

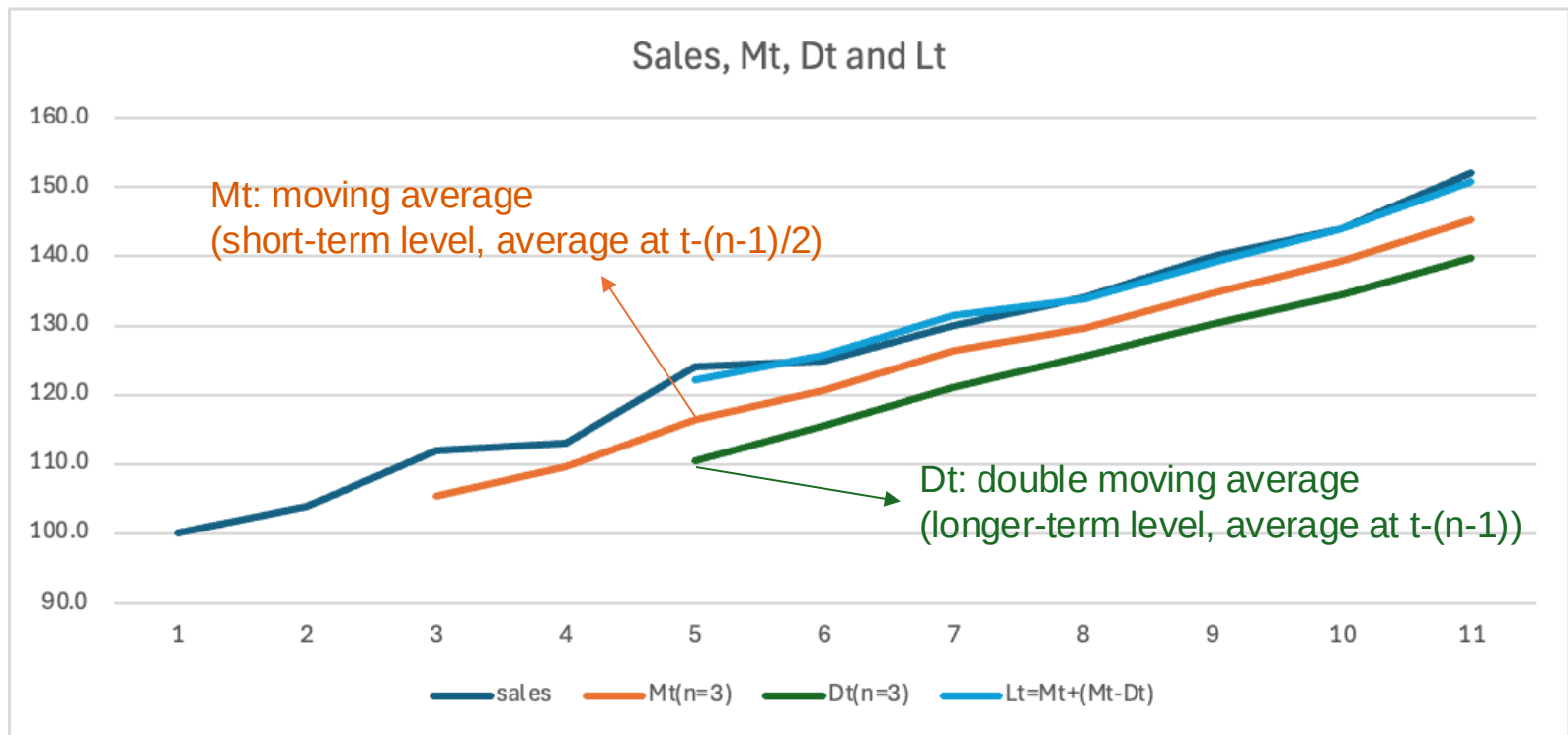
Why DMV works?

- ❖ An illustration with a linear sales prediction case (see Why Double Moving Average Works.csv)



Why DMV works?

- ❖ An illustration with a linear sales prediction case (see Why Double Moving Average Works.csv)



Why DMV works?

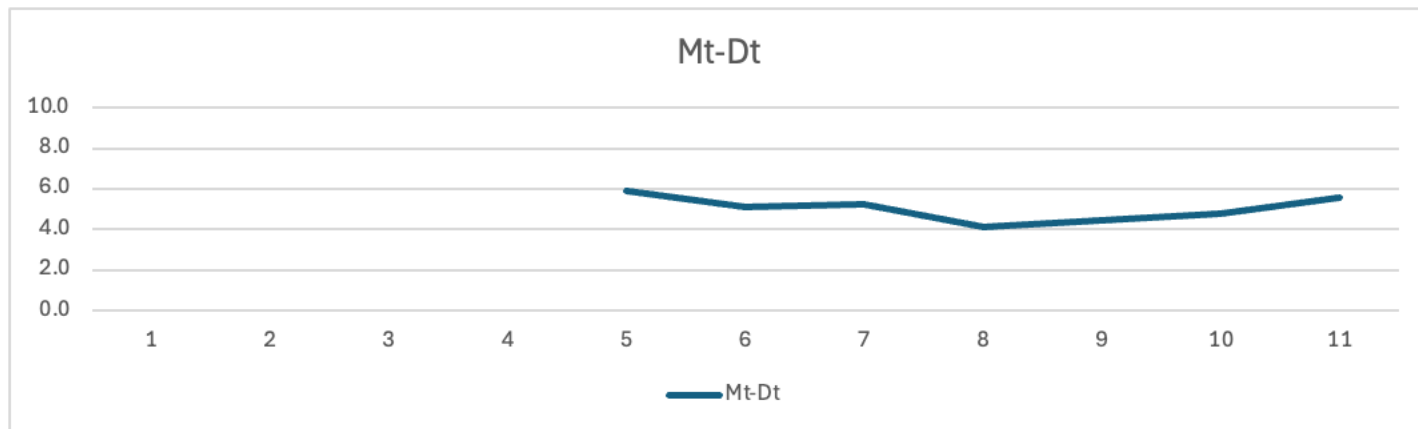
- ❖ An illustration with a linear sales prediction case
(see Why Double Moving Average Works.csv)

Mt: moving average
(short-term level, average at $t-(n-1)/2$)

The difference between Mt and Dt:
How sales go up for time duration $(n-1)/2$

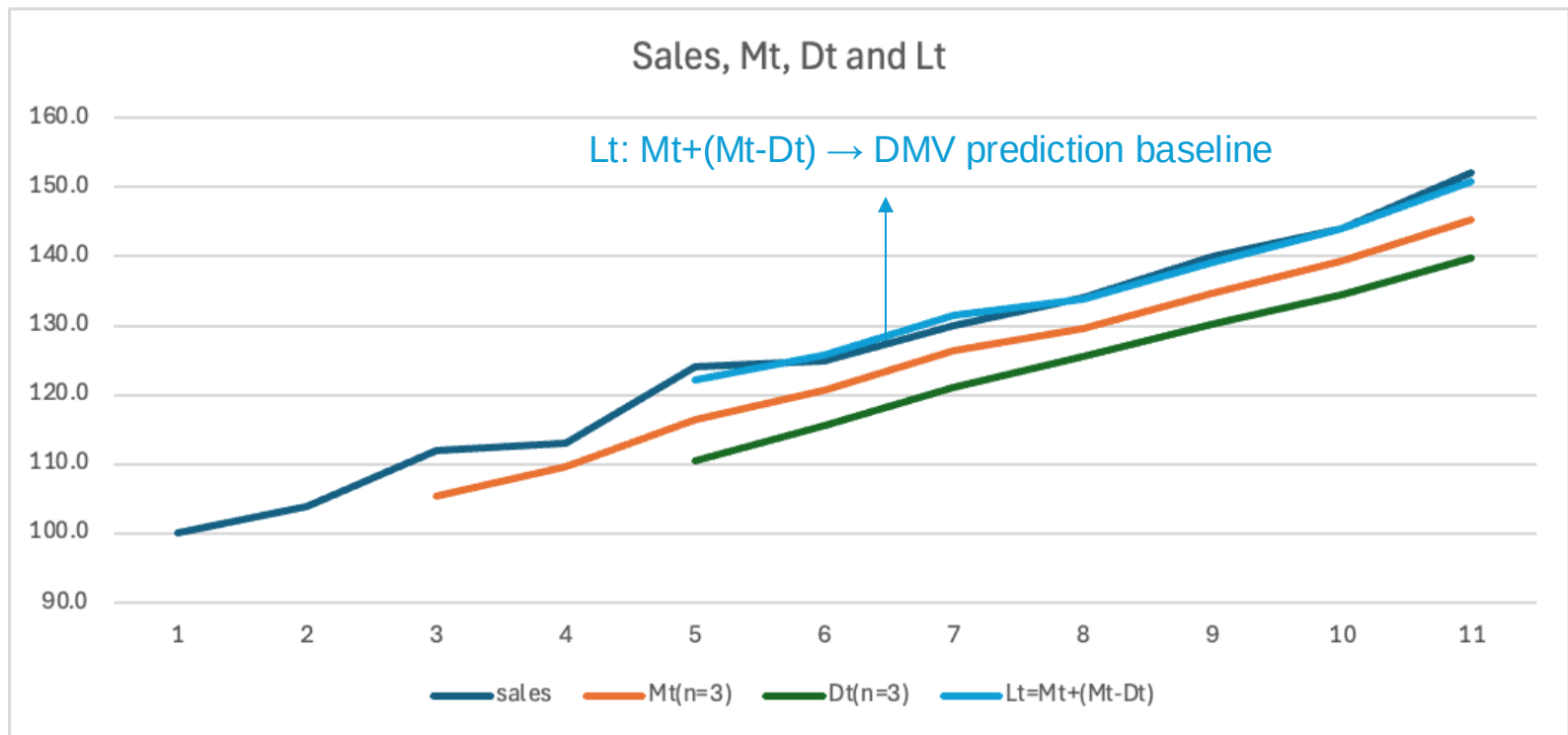
Dt: double moving average
(longer-term level, average at $t-(n-1)$)

We can use this to adjust the Mt, then adjusted Mt serves as the prediction baseline



Why DMV works?

- ❖ An illustration with a linear sales prediction case (see Why Double Moving Average Works.csv)



Holt-Linear

Simple exponential smoothing + trend

- When there is a trend in the series, Holt's method deals with it explicitly by including a trend term, T_t , and a corresponding smoothing constant $\mathcal{B}$.
- L_t : The level of time series.
- T_t : An estimate of the change in the series from one period to the next.
- Formulas for Holt's exponential smoothing method:

$$L_t = \alpha Y_t + (1 - \alpha)(L_{t-1} + T_{t-1}) \rightarrow F_t$$

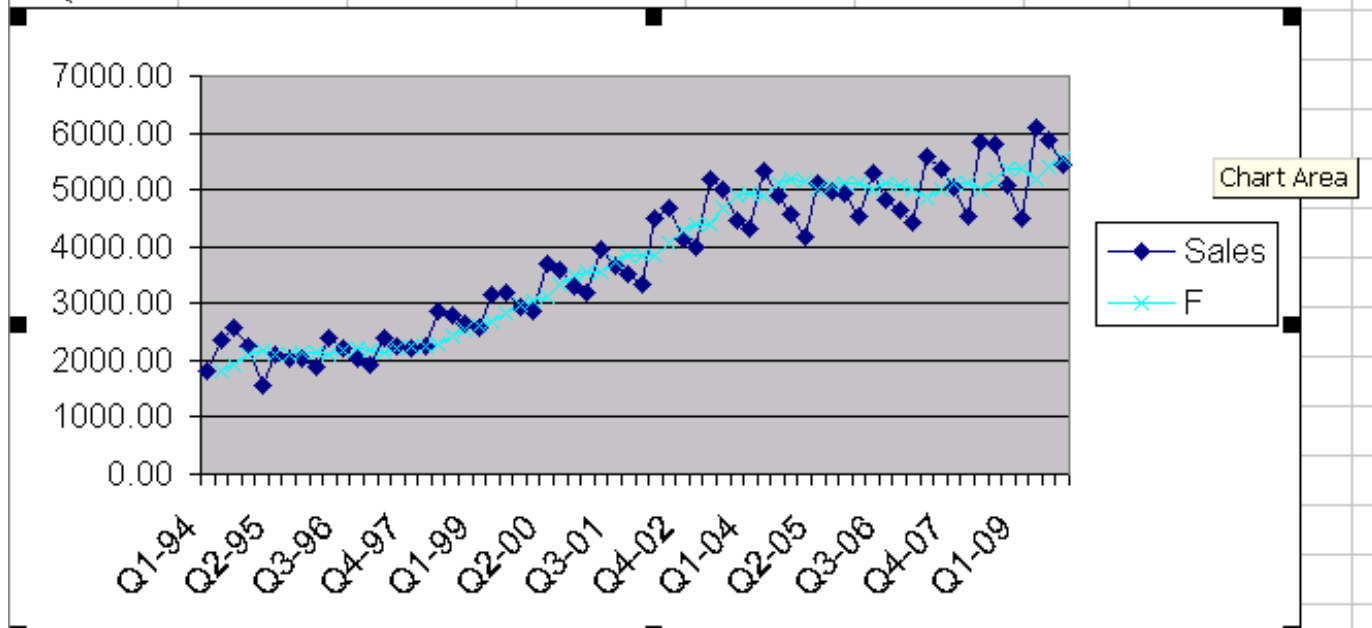
$$T_t = \beta(L_t - L_{t-1}) + (1 - \beta)T_{t-1}$$

$$F_{t+k} = L_t + kT_t$$

Increase in the trend

Soft Drink Sales

Quarter	Sales	L	T	F	e	Alpha	0.193338836
Q1-94	1807.37	1807.37	0			Beta	0.195071241
Q2-94	2355.32	1913.31	20.66585	1807.37	547.95	RMSE	440.1662445
Q3-94	2591.83	2061.165	45.47672	1933.976	657.85		
Q4-94	2236.39	2131.727	50.37017	2106.641	129.75		
Q1-95	1549.14	2059.722	26.49829	2182.097	-632.96		



Holt-Winter

Simple exponential smoothing + trend + seasonality

- Current level (l_t), similar to the Holt-Linear
- Current trend (b_t), similar to the Holt-Linear
- Current seasonality (s_t)

Holt-Winter

- Similar to Holt's model, but also has seasonal indexes and a corresponding smoothing constant γ (gamma).
- The smoothing constant gamma controls how quickly the method reacts to observed changes in the seasonality pattern.

$$L_t = \alpha \frac{Y_t}{S_{t-M}} + (1-\alpha)(L_{t-1} + T_{t-1})$$

Trying to estimate Y/S

$$T_t = \beta(L_t - L_{t-1}) + (1-\beta)T_{t-1}$$

Seasonal index

$$S_t = \gamma \frac{Y_t}{L_t} + (1-\gamma)S_{t-M}$$

Trying to estimate S using Y/L

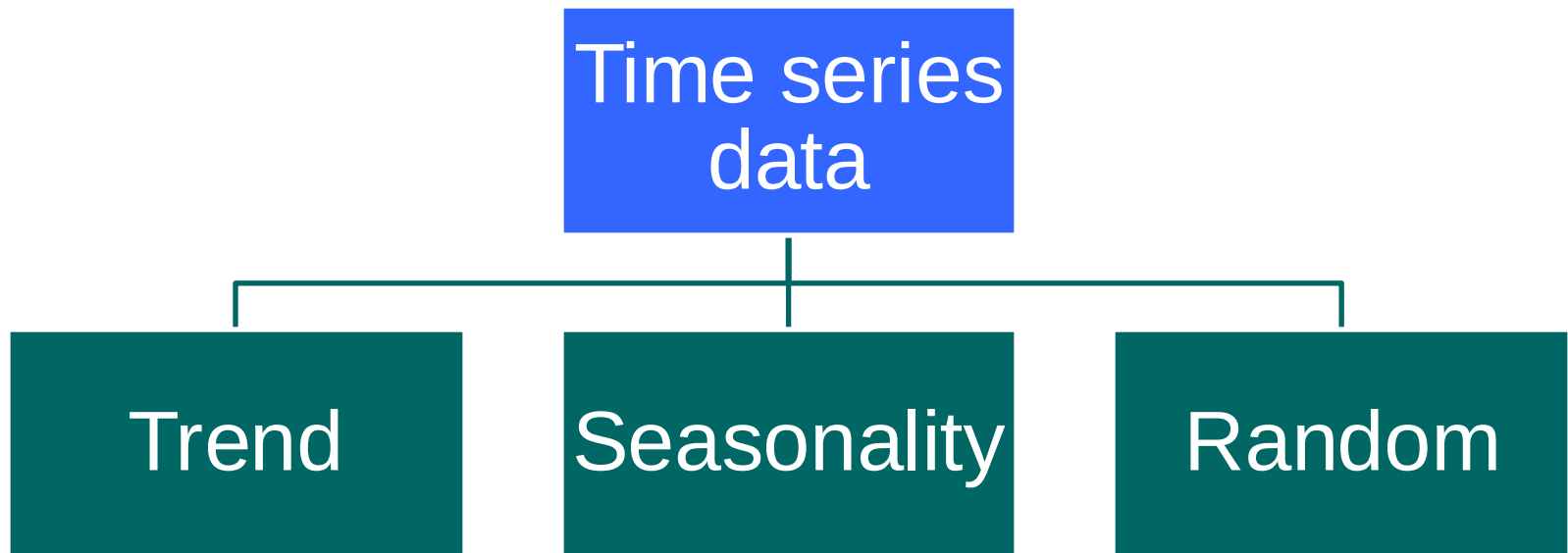
of season

$$F_{t+k} = (L_t + kT_t)S_{t+k-M}$$

Additive trend times
multiplicative seasonal effect

ETS

ETS: error, trend, seasonality



Additive or multiplicative?

ETS (error, trend, seasonality)

Error Component	Trend Component	Seasonal Component		
		None	Additive	Multiplicative
Additive	None	A, N, N	A, N, A	Not Used
	Additive	A, A, N	A, A, A	
	Damped	A, A _d , N	A, A _d , A	
Multiplicative	None	M, N, N	M, N, A	M, N, M
	Additive	M, A, N	M, A, A	M, A, M
	Damped	M, A _d , N	M, A _d , A	M, A _d , M

- ✓ *SES: ETS(A, N, N)*
- ✓ *Holt linear: ETS(A, A, N) and damped is ETS(A, A_d, N)*
- ✓ *Holt Winter: ETS(A, A, A) and damped is ETS(A, A_d, A)*

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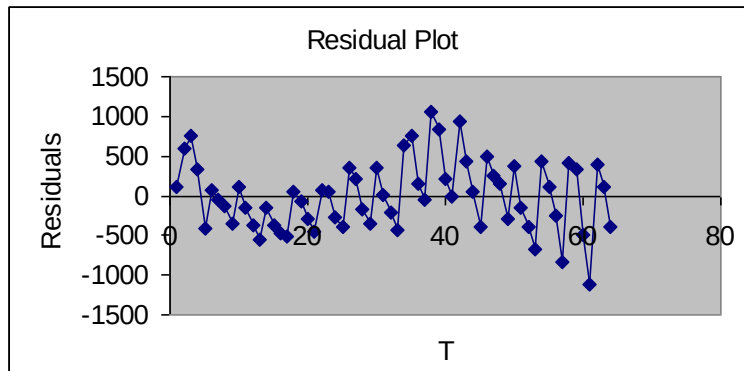
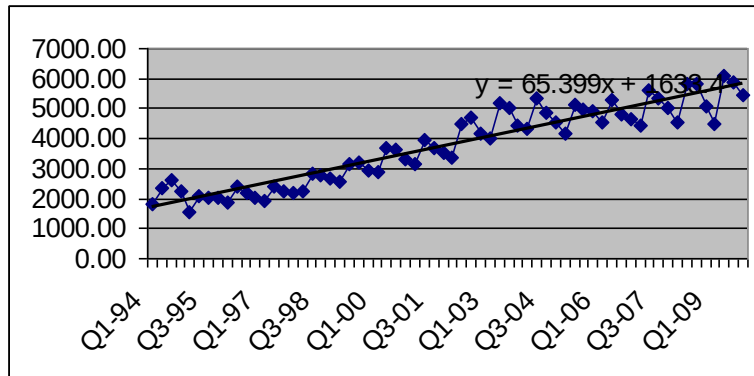
- ☐ Introduction to time series
- ☐ Quantitative Forecasting
 - ☐ Statistical Methods
 - ☒ **Regression Analysis**

Regression on Linear Trend

- ❖ A linear trend means that the time series variable changes by a constant amount each time period.
- ❖ Linear trend model is given by: $Y_t = a + bt + e_t$
- ❖ Interpretation of ***b*** is that it represents the expected change in the series from one period to the next.
 - ⌘ If *b* is positive, the trend is upward.
 - ⌘ If *b* is negative, the trend is downward.
- ❖ The intercept term *a* is less important: It literally represents the expected value of the series at time 0.
- ❖ As always, a graph of the time series is a good place to start. It may indicate whether a linear trend is likely to provide a good fit.

Soft Drink Sales

❖ $\text{Sales}_t = a + b \cdot t + e_t$



SUMMARY OUTPUT						
<i>Regression Statistics</i>						
Multiple R	0.941304					
R Square	0.886054					
Adjusted R	0.884216					
Standard Error	440.1709					
Observations	64					
ANOVA						
	<i>df</i>	<i>SS</i>	<i>MS</i>	<i>F</i>	<i>Significance F</i>	
Regression	1	93410374	93410374	482.117	6.12E-31	
Residual	62	12012527	193750.4			
Total	63	1.05E+08				
	<i>Coefficients</i>	<i>Standard Error</i>	<i>t Stat</i>	<i>P-value</i>	<i>Lower 95%</i>	<i>Upper 95%</i>
Intercept	1633.356	111.3451	14.66931	7.85E-22	1410.78	1855.931
T	65.39902	2.978482	21.95716	6.12E-31	59.44512	71.35292

Regression on Exponential Trend

- ❖ An exponential trend is appropriate when the time series changes by a constant percentage (as opposed to a constant amount) each period.
- ❖ Appropriate regression equation contains a multiplicative error term u_t : $Y_t = ce^{bt}u_t$.
- ❖ **Exponential trend is not useful for estimation, and a linear equation is required.** You can achieve linearity by taking natural logarithms of both sides of the equation: $\ln(Y_t) = a + bt + e_t$.
- ❖ Interpreting final result: The coefficient b (expressed as a percentage) is approximately the percentage change per period. For example, if $b = 0.05$, then the series is increasing by approximately 5% per period.

Regression on Seasonal Effect

- A regression approach to forecasting seasonal data **uses dummy variables for the seasons.**
 - For quarters, create three dummy variables, and the coefficients of the dummy variables, b_1 , b_2 , and b_3 , indicate how much each quarter differs from the reference quarter.
- Depending on how the regression equations are written, you can create either an additive or a multiplicative seasonal model.

Soft Drink Sales

- Consider both trend and seasonal effect
- $\text{LogSales} = b_0 + b_1t + b_2Q_1 + b_3Q_2 + b_4Q_3$
 - The figure illustrates the data setup, with dummy variables for quarters and a transformed *sales* ($\text{Log}(\text{sales})$) variable.
 - Regression is straightforward.

	A	B	C	D	E	F	G
1	Quarter	Sales	Time	Q1	Q2	Q3	Log(Sales)
2	Q1-94	1807.37	1	1	0	0	7.499628
3	Q2-94	2355.32	2	0	1	0	7.7644319
4	Q3-94	2591.83	3	0	0	1	7.8601195
5	Q4-94	2236.39	4	0	0	0	7.7126187
6	Q1-95	1549.14	5	1	0	0	7.3454
7	Q2-95	2105.79	6	0	1	0	7.652
8	Q3-95	2041.32	7	0	0	1	7.6213
9	Q4-95	2021.01	8	0	0	0	7.6113
10	Q1-96	1870.46	9	1	0	0	7.5339
11	Q2-96	2390.56	10	0	1	0	7.7792
12	Q3-96	2198.03	11	0	0	1	7.6953
13	Q4-96	2046.83	12	0	0	0	7.6240
14	Q1-97	1934.19	13	1	0	0	7.5674
15	Q2-97	2406.41	14	0	1	0	7.7858

	A	B	C	D	E	F	G
7		Multiple	R-Square	Adjusted	SE of		
8	Summary	R		R-Square	Estimate		
9		0.9628	0.9270	0.9218	0.102		
10		Degrees of	Sum of	Mean of	F-Ratio	p-Value	
11	ANOVA Table	Freedom	Squares	Squares			
12		4	7.465	1.866	177.8172	< 0.0001	
13	Explained						
14	Unexplained	56	0.588	0.010			
15		Coefficient	Standard	t-Value	p-Value	Confidence Interval 95%	
16	Regression Table		Error			Lower	Upper
17		7.510	0.036	210.8236	< 0.0001	7.439	7.581
18	Constant						
19	Time	0.019	0.001	25.9232	< 0.0001	0.018	0.021
20	Q1	-0.090	0.037	-2.4548	0.0172	-0.164	-0.017
21	Q2	0.140	0.037	3.7289	0.0005	0.065	0.215
22	Q3	0.091	0.037	2.4449	0.0177	0.017	0.166

How to model this lagged features in sklearn?

- ❖ https://scikit-learn.org/stable/auto_examples/applications/plot_time_series_lagged_features.html

```
lagged_df = df.select(
    "count",
    *[pl.col("count").shift(i).alias(f"lagged_count_{i}h") for i in [1, 2, 3]],
    lagged_count_1d=pl.col("count").shift(24),
    lagged_count_1d_1h=pl.col("count").shift(24 + 1),
    lagged_count_7d=pl.col("count").shift(7 * 24),
    lagged_count_7d_1h=pl.col("count").shift(7 * 24 + 1),
    lagged_mean_24h=pl.col("count").shift(1).rolling_mean(24),
    lagged_max_24h=pl.col("count").shift(1).rolling_max(24),
    lagged_min_24h=pl.col("count").shift(1).rolling_min(24),
    lagged_mean_7d=pl.col("count").shift(1).rolling_mean(7 * 24),
    lagged_max_7d=pl.col("count").shift(1).rolling_max(7 * 24),
    lagged_min_7d=pl.col("count").shift(1).rolling_min(7 * 24),
)
lagged_df.tail(10)
```

month	sales	shift(1)	shift(1).rolling_mean(3)	shift(1).rolling_max(3)	shift(1).rolling_min(3)
1	100				
2	104	100			
3	112	104			
4	113	112	=AVERAGE(B2:B4)	112	100

(see model lagged term in sklearn.csv)

Want to model other time-related features in sklearn?

- ❖ https://scikit-learn.org/stable/auto_examples/applications/plot_cyclical_feature_engineering.html

e.g., one-hot encoder

```
one_hot_linear_pipeline = make_pipeline(
    ColumnTransformer(
        transformers=[
            ("categorical", one_hot_encoder, categorical_columns),
            ("one_hot_time", one_hot_encoder, ["hour", "weekday", "month"]),
        ],
        remainder=MinMaxScaler(),
    ),
    RidgeCV(alphas=alphas),
)

evaluate(one_hot_linear_pipeline, X, y, cv=ts_cv)
```